

The volume of a prism

Base area times height, for every prism — and the dissection that proves it.

LESSON 26–27

2 × 40 MIN

GEOMETRY 11, §13

IGCSE E13.X

LESSON CLOCK

14'

20'

20'

22'

16'

8'

From the cuboid to any prism	14 min
Applying the formula	20 min
Working backwards	20 min
Compound solids	22 min
Practice	16 min
Homework	8 min

LEARNING OBJECTIVES

- Derive $V = Bh$ for a triangular prism from the parallelepiped.
- Compute the volume of any prism.
- Solve for a missing dimension from a given volume.
- Handle compound solids made of prisms.

TEXTBOOK

National	pp. 129–140
Cambridge	Core & Extended, pp. 305–310
Workbook	Exercise 13.1

01

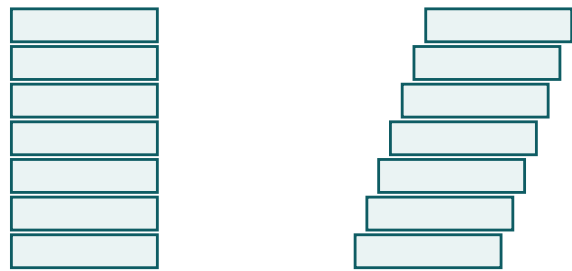
Explanation

From the cuboid to any prism

- Step 1.** A parallelepiped is a cuboid with the layers slid sideways, so $V = Bh$ by Cavalieri.
- Step 2.** A triangular prism is half a parallelepiped, cut along a diagonal plane. Both halves are congruent, so each has volume $\frac{1}{2}B_p h = Bh$, where B is now the triangle's area.
- Step 3.** Any polygonal base splits into triangles; the prism splits into triangular prisms of the same height, and the volumes add.

$$V = B \cdot h \text{ for every prism}$$

same base, same height, same volume



right prism oblique prism
every horizontal slice has the same area

Cavalieri's principle: equal slices at every height mean equal volumes.

THE IDEA BEHIND EVERY VOLUME FORMULA

A solid of uniform cross-section has volume = area of cross-section × length. The cylinder of the next lesson is the same statement with a circle instead of a polygon.

Applying it

PRISM	B	V
cuboid $a \times b$, height h	ab	abh
right triangular, legs p, q	$\frac{1}{2}pq$	$\frac{1}{2}pqh$
regular triangular, edge a	$\frac{\sqrt{3}}{4}a^2$	$\frac{\sqrt{3}}{4}a^2h$
regular hexagonal, edge a	$\frac{3\sqrt{3}}{2}a^2$	$\frac{3\sqrt{3}}{2}a^2h$
trapezium base p, q , depth d	$\frac{p+q}{2}d$	$\frac{p+q}{2}dh$

Compound solids

Real objects — an L-shaped room, a channel, a house with a roof — are prisms with a compound cross-section. Find the area of the whole cross-section first, then multiply once.

Example. A shed is a cuboid 6×4 m and 2.5 m to the eaves, with a triangular roof 1.5 m high. Cross-section: rectangle $4 \times 2.5 = 10$ plus triangle $\frac{1}{2} \times 4 \times 1.5 = 3$, total 13 m^2 . Volume $13 \times 6 = 78 \text{ m}^3$.

MULTIPLY ONCE, AT THE END

Computing each part's volume separately works too, but doubles the arithmetic and the chances of error. Add the areas first.

02 Worked examples

EXAMPLE 1 A regular hexagonal prism has base edge 5 cm and height 12 cm. Find its volume.

$$\textcircled{1} \quad B = \frac{3\sqrt{3}}{2} \times 25 \approx 64.95$$

$$\textcircled{2} \quad V = 64.95 \times 12$$

$$\textcircled{3} \quad \approx 779.4$$

ANSWER

$$150\sqrt{3} \times \frac{12}{2} \text{ — i.e. } \approx 779 \text{ cm}^3$$

EXAMPLE 2 A trough has a trapezium cross-section with parallel sides 40 cm and 60 cm, depth 30 cm, and is 2 m long. Find its capacity in litres.

$$\textcircled{1} \quad B = \frac{40 + 60}{2} \times 30 = 1500 \text{ cm}^2.$$

$$\textcircled{2} \quad V = 1500 \times 200 = 300\,000 \text{ cm}^3.$$

$$\textcircled{3} \quad = 300 \text{ litres.}$$

ANSWER

300 litres

EXAMPLE 3 A prism has volume 540 cm³ and a triangular base with legs 6 and 9. Find its height.

$$\textcircled{1} \quad B = \frac{1}{2} \times 6 \times 9 = 27$$

$$\textcircled{2} \quad 27h = 540$$

$$\textcircled{3} \quad h = 20$$

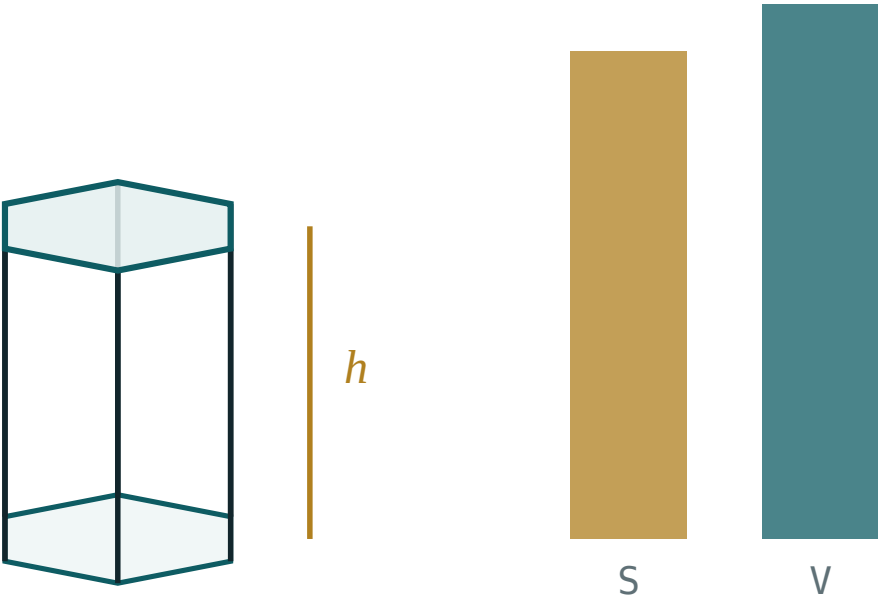
ANSWER

20 cm

03 Interactive model

Cut a parallelepiped of plasticine along a diagonal plane and compare the two halves.

The volume of a prism



base area B **41.6**

perimeter P **24**

total S **227.1**

volume V **249.4**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Volume of a prism	Prizma hajmi	Объём призмы
Base area	Asos yuzasi	Площадь основания
Height	Balandlik	Высота
Dissection	Bo'lish	Разбиение
Cavalieri's principle	Kavaleri prinsipi	Принцип Кавальери
Compound solid	Murakkab jism	Составное тело
Cross-section	Ko'ndalang kesim	Поперечное сечение
Uniform cross-section	O'zgarmas kesim	Постоянное сечение
Capacity	Sig'im	Вместимость

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The volume of any prism is:

Ph

Bh

$2B + Ph$

$\frac{1}{3} Bh$

A triangular prism is:

a third of a cuboid

half a parallelepiped

a quarter of a cube

unrelated

Cavalieri's principle says:

equal bases give equal volumes

equal slice areas at every height give equal volumes

volumes always add

nothing about volume

A prism of $B = 27$ and $V = 540$ has height:

15

20

27

30

For a compound cross-section you should:

multiply each part

add the areas then multiply once

take the largest

average them

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Volume with $B = 12$, $h = 5$

ANSWER 60

2. Cuboid $4 \times 5 \times 6$

ANSWER 120

3. Right triangular prism, legs 3 and 4, height 10

ANSWER 60

4. Cube of edge 8

ANSWER 512

5. Prism $B = 25$, $h = 4$

ANSWER 100

6. A prism of $V = 100$, $B = 20$: the height

ANSWER 5

7. Regular triangular prism, edge 4, height 6

ANSWER $24\sqrt{3}$

Medium · the standard question

7 PROBLEMS

1. Regular hexagonal prism, edge 5, height 12

ANSWER $\approx 779 \text{ cm}^3$

2. Trough, trapezium 40/60 by 30 cm, 2 m long: litres

ANSWER 300

3. Prism $V = 540$, triangular base legs 6 and 9: height

ANSWER 20

4. Regular hexagonal prism, edge 4, height 10

ANSWER $240\sqrt{3} \approx 416$

5. Shed: rectangle 4×2.5 plus roof triangle height 1.5, length 6 m

ANSWER 78 m^3

6. An L-shaped cross-section of area 34 cm^2 , length 15 cm

ANSWER 510 cm^3

7. A cuboid of volume 600 with base 10×12 : height

ANSWER 5

Hard · stretch and reason

7 PROBLEMS

1. A prism has a rhombus base with diagonals 10 and 24, height 15. Volume

ANSWER 1800

2. A regular hexagonal prism of volume $648\sqrt{3}$ and edge 6: its height

ANSWER 12

3. A channel of trapezium section 1.2 m and 2 m wide, 0.8 m deep, 30 m long: capacity in m^3

ANSWER 38.4

4. Two similar prisms of volumes 54 and 128: the surface-area ratio

ANSWER 9 : 16

5. A cube is cut by a plane through A , C , B_1 . Find the volume of the smaller piece for edge 6

ANSWER 36

6. A prism's height doubles and its base edges halve. What happens to the volume?

ANSWER It halves

7. Prove $V = Bh$ for a triangular prism

ANSWER It is half a parallelepiped on twice the base

07 Homework

Homework — 5 tasks

Add the cross-sectional areas before multiplying by the length.

1. A regular hexagonal prism has base edge 6 cm and height 14 cm. Find its volume.
2. A trough has a trapezium section with parallel sides 30 cm and 50 cm and depth 25 cm, and is 3 m long. Find its capacity in litres.
3. A prism has volume 420 cm^3 and a triangular base with legs 7 and 8. Find its height.

4. A shed has a rectangular cross-section 5×3 m with a triangular roof 2 m high, and is 8 m long. Find its volume.
5. Prove that a triangular prism has volume Bh , starting from the parallelepiped.